SRI SHANMUGHA
COLLEGE OF ENGINEERING AND TECHNOLOGY
(Autonomous)

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AI & DATA SCIENCE

RECORD NOTEBOOK

23CS412 – Embedded System and IoT

2026-2027

Semester-Even

REGULATION - 2023

NAMEOFTHESTUDENT	:	
REGISTERNUMBER	:	
YEAR/SEM	:	
COURSE/BRANCH	:	

SRI SHANMUGHA
COLLEGE OF ENGINEERING AND TECHNOLOGY
(Autonomous)

RECORD NOTEBOOK

REGISTERNUMBER : ……………………..…………………

Certified that this is a Bonafide record of Practical work done by

Mr./Ms.. ....of the **{semester}**

Semester

... .Branch during the Academic year

R 2023 Even Semester in the **23CS412 Embedded System and IoT** Laboratory.

Signature of Staff In-charge **Signature of Head of Department**

VISION AND MISSION OF THE INSTITUTE

VISION

To be an Institute of repute in the field of Engineering and Technology by implementing the best educational practices akin to global standards for fostering domain knowledge and developing research attitude among students to make them globally competent.

MISSION

- AchievingexcellenceinTeachingLearningprocessusingstate-of-the-art resources
- Extending opportunity to upgrade faculty knowledge and skills.
- Implementing the best student training practices for requirements of industrial scenario of the state
- Motivatingfacultyandstudentsinresearchactivityforrealtimeapplication

VISION AND MISSION OF THE DEPARTMENT

VISION

To create the holistic environment for the development of Computer Science and Engineering Graduates employable at the global level and to mold them through comprehensive educational programs and quality research for developing their competency and innovation with moral values

MISSION

M1: Ensuring academic growth by way of establishing centers of excellence and promoting collaborative learning

M2: Promoting research-based projects in the emerging areas of technology convergence for the benefit of students and faculty

M3: Motivating the students to be successful, ethical and suitable for industry ready

Program Outcomes(POs):

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health, safety, cultural, societal and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis, and interpretation of data and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, apply appropriate techniques, resources, modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal, environmental contexts, demonstrate the knowledge and need for sustainable development.

PO8: Ethics: Apply ethical principles, commit to professional ethics, responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, as a member or leader in diverse teams and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community with society at large being able to comprehend, write effective reports, design documentation, make effective presentations and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge, understandingoftheengineeringandmanagementprinciplesandapplythese to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12:Life-longlearning: Recognizetheneed,abilitytoengageinindependent and life-long learning in the broadest context of technological change.

Program Specific Objectives(PSOs)

PSO1: Ability to apply programming skills for solving real time problems in the areas related to algorithms, data structures, cloud computing and data science.

PSO2: Ability to develop high quality software products using cutting edge technologies.

Program Educational Objective(PEOs)

PEO1 Basic Skills: Graduates work productively as successful Computer Professionals with problem solving skills, core computing skills and soft skills with social awareness

PEO2 Technical Knowledge: Graduates engage in everlasting endeavor to promote research and development

PEO3 Managerial Skills: Graduates communicate effectively, recognize and incorporate the societal needs in their profession by practicing their boundless skills with high regard to ethical responsibilities

TABLE OF CONTENTS

EX NO: 1	Experiment 1: IoT Sensor Data Acquisition using Raspberry Pi
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 1 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:

GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?

A1: A low-cost, credit-card sized single-board computer.

Q2: What does DHT11 measure?

A2: Temperature and humidity.

Q3: What is GPIO?

A3: General Purpose Input/Output pins for interfacing hardware.

Q4: What is IoT?

A4: Internet of Things - network of physical devices connected via internet.

Q5: What voltage does Raspberry Pi operate on?

A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO1 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 2	Experiment 2: IoT Sensor Data Acquisition using Raspberry Pi
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 2 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:

GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?

A1: A low-cost, credit-card sized single-board computer.

Q2: What does DHT11 measure?

A2: Temperature and humidity.

Q3: What is GPIO?

A3: General Purpose Input/Output pins for interfacing hardware.

Q4: What is IoT?

A4: Internet of Things - network of physical devices connected via internet.

Q5: What voltage does Raspberry Pi operate on?

A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO2 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 3	Experiment 3: IoT Sensor Data Acquisition using Raspberry Pi
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 3 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:
GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?
A1: A low-cost, credit-card sized single-board computer.
Q2: What does DHT11 measure?
A2: Temperature and humidity.
Q3: What is GPIO?
A3: General Purpose Input/Output pins for interfacing hardware.
Q4: What is IoT?
A4: Internet of Things - network of physical devices connected via internet.
Q5: What voltage does Raspberry Pi operate on?
A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO3 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 4	Experiment 4: IoT Sensor Data Acquisition using Raspberry Pi
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 4 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:

GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?

A1: A low-cost, credit-card sized single-board computer.

Q2: What does DHT11 measure?

A2: Temperature and humidity.

Q3: What is GPIO?

A3: General Purpose Input/Output pins for interfacing hardware.

Q4: What is IoT?

A4: Internet of Things - network of physical devices connected via internet.

Q5: What voltage does Raspberry Pi operate on?

A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO4 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 5	**Experiment 5: IoT Sensor Data Acquisition using Raspberry Pi**
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 5 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:

GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?

A1: A low-cost, credit-card sized single-board computer.

Q2: What does DHT11 measure?

A2: Temperature and humidity.

Q3: What is GPIO?

A3: General Purpose Input/Output pins for interfacing hardware.

Q4: What is IoT?

A4: Internet of Things - network of physical devices connected via internet.

Q5: What voltage does Raspberry Pi operate on?

A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO5 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 6	**Experiment 6: IoT Sensor Data Acquisition using Raspberry Pi**
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 6 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:
GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?
A1: A low-cost, credit-card sized single-board computer.
Q2: What does DHT11 measure?
A2: Temperature and humidity.
Q3: What is GPIO?
A3: General Purpose Input/Output pins for interfacing hardware.
Q4: What is IoT?
A4: Internet of Things - network of physical devices connected via internet.
Q5: What voltage does Raspberry Pi operate on?
A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO5 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 7	**Experiment 7: IoT Sensor Data Acquisition using Raspberry Pi**
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 7 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:
GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?
A1: A low-cost, credit-card sized single-board computer.
Q2: What does DHT11 measure?
A2: Temperature and humidity.
Q3: What is GPIO?
A3: General Purpose Input/Output pins for interfacing hardware.
Q4: What is IoT?
A4: Internet of Things - network of physical devices connected via internet.
Q5: What voltage does Raspberry Pi operate on?
A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO5 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 8	Experiment 8: IoT Sensor Data Acquisition using Raspberry Pi
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 8 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:

GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?

A1: A low-cost, credit-card sized single-board computer.

Q2: What does DHT11 measure?

A2: Temperature and humidity.

Q3: What is GPIO?

A3: General Purpose Input/Output pins for interfacing hardware.

Q4: What is IoT?

A4: Internet of Things - network of physical devices connected via internet.

Q5: What voltage does Raspberry Pi operate on?

A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO5 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 9	**Experiment 9: IoT Sensor Data Acquisition using Raspberry Pi**
DATE:	

AIM:

To design and implement a sensor data acquisition system for experiment 9 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:

GPIO4 = DHT11 sensor data pin

Sample Output:
Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?

A1: A low-cost, credit-card sized single-board computer.

Q2: What does DHT11 measure?

A2: Temperature and humidity.

Q3: What is GPIO?

A3: General Purpose Input/Output pins for interfacing hardware.

Q4: What is IoT?

A4: Internet of Things - network of physical devices connected via internet.

Q5: What voltage does Raspberry Pi operate on?

A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO5 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

EX NO: 10 Experiment 10: IoT Sensor Data Acquisition using Raspberry Pi

AIM:

To design and implement a sensor data acquisition system for experiment 10 using Raspberry Pi and IoT sensors.

OBJECTIVES:

1. Understand the basics of Raspberry Pi GPIO programming
2. Interface DHT11 sensor with Raspberry Pi
3. Read temperature and humidity data in real-time
4. Display sensor data on LCD module

THEORY:

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a monitor or TV. IoT (Internet of Things) enables physical objects to connect and exchange data. The DHT11 sensor measures temperature (0-50°C) and humidity (20-90% RH).

REQUIREMENTS:

Hardware: Raspberry Pi 3B+, DHT11 Sensor, Breadboard, Jumper wires

Software: Raspbian OS, Python 3.x, RPi.GPIO library

ALGORITHM:

1. Initialize GPIO pins
2. Configure DHT11 sensor pin
3. Read sensor data
4. Parse temperature and humidity values
5. Display on LCD

PROCEDURE:

1. Connect DHT11 sensor DATA pin to GPIO4
2. Connect VCC to 3.3V and GND to ground
3. Install RPi.GPIO and Adafruit_DHT libraries
4. Write Python script to read sensor values
5. Run the program and observe output

PROGRAM:

```
import RPi.GPIO as GPIO
import Adafruit_DHT

SENSOR = Adafruit_DHT.DHT11
PIN = 4

humidity, temperature = Adafruit_DHT.read_retry(SENSOR, PIN)
if humidity and temperature:
print(f'Temp={temperature:.1f}C  Humidity={humidity:.1f}%')
else:
print('Sensor read failed')
```

OBSERVATIONS & OUTPUT:

Sample Input:

GPIO4 = DHT11 sensor data pin

Sample Output:

Temp=28.0C Humidity=65.0%

RESULT:

The experiment was successfully completed. Temperature and humidity data were acquired from DHT11 sensor and displayed correctly.

VIVA QUESTIONS:

Q1: What is Raspberry Pi?

A1: A low-cost, credit-card sized single-board computer.

Q2: What does DHT11 measure?

A2: Temperature and humidity.

Q3: What is GPIO?

A3: General Purpose Input/Output pins for interfacing hardware.

Q4: What is IoT?

A4: Internet of Things - network of physical devices connected via internet.

Q5: What voltage does Raspberry Pi operate on?

A5: 3.3V and 5V.

MAPPING:

CO Mapping: CO5 | Bloom Level: K3 - Apply

REFERENCES:

1. T. Hughes, 'Raspberry Pi User Guide', Wiley, 2016.

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